Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Project Detail	Value
Date	21 October 2026
Team ID	Individual Project
Project Name	StaffSync – Modern Human Resource Management System
Maximum Marks	4 Marks

Functional Requirements

The following functional requirements describe the main capabilities expected from StaffSync, a MERN-stack Human Resource Management System with Employee, Manager, and HR roles.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Access Control	Employee registration and login; secure password hashing; JWT-based authentication; protected routes; role-based access for Employee, Manager, and HR; validation of authentication inputs.
FR-2	Employee & Manager Management	HR can create, view, and update employee records and manage manager accounts. Employees can view and update permitted profile details. Manager access is limited to authorized team information.
FR-3	Attendance Management	Employees can record attendance and view attendance history. HR can review organization-wide attendance records; managers can review attendance for their authorized teams.
FR-4	Leave Management & Approval	Employees can submit leave requests and track their status. HR and authorized managers can review, approve, or reject requests according to their permissions.
FR-5	Announcements & Notifications	HR can publish announcements. The system can provide user-facing notifications for relevant actions such as leave decisions and announcements.
FR-6	Training Management	Authorized users can manage training information, and employees can view available training programs and related details, subject to implemented permissions.
FR-7	Performance Management	Authorized users can create or view performance records and feedback according to role permissions.
FR-8	Dashboard & Reporting	Provide role-appropriate dashboard summaries for employee information, attendance, leave status, and other available HRMS metrics.
FR-9	Data Management & Validation	Validate incoming data, maintain relationships between user and HR records, prevent invalid or unauthorized operations, and return meaningful API responses.

Non-functional Requirements

The following quality requirements describe expected system characteristics. Quantitative targets should be verified through testing rather than treated as measured results.

NFR No.	Non-functional Requirement	Description
NFR-1	Usability	The interface should be clear and consistent for Employees, Managers, and HR, with understandable navigation, form validation, and feedback for successful or failed actions.
NFR-2	Security	Passwords must be hashed; authentication should use JWTs; server-side role-based authorization must protect sensitive operations; secrets must be stored in environment variables; inputs should be validated; unauthorized requests must be rejected.
NFR-3	Reliability	The system should handle database or API errors gracefully, preserve valid records, avoid duplicate or inconsistent attendance/leave operations, and return predictable error responses.
NFR-4	Performance	Common pages and API operations should respond promptly under expected project-demo load. Establish and measure concrete response-time targets using tests before claiming specific timings.
NFR-5	Availability	The deployed application should recover from service interruptions and expose health checks for operational monitoring. Uptime targets must be confirmed from deployment requirements and measured monitoring.
NFR-6	Scalability	Use pagination for large lists, database indexes for frequently queried fields, and efficient API queries. The architecture should permit future scaling without claiming a microservices design unless it is actually implemented.
NFR-7	Maintainability	Keep routes, controllers, models, middleware, and frontend services organized; use consistent naming, environment configuration, clear error handling, and automated tests.
NFR-8	Compatibility	The web application should work in current mainstream browsers and provide a usable layout across common desktop and mobile screen sizes, subject to responsive testing.
NFR-9	Data Privacy & Integrity	Only authorized roles should access sensitive employee data. Updates should be validated, and personal or authentication data should not be unnecessarily exposed in API responses or logs.

Acceptance and Verification Notes

- Verify that public registration cannot grant HR or Manager privileges.
- Test employee, manager, and HR permissions independently for protected API endpoints.
- Confirm password hashes are stored rather than plaintext passwords.
- Test leave approval, attendance recording, notifications, and dashboard summaries against database records.
- Run frontend build and backend/API tests before submission.
- Treat performance and availability statements as targets until they have been measured in the deployed environment.